

Research

Multi-criteria decision analysis for regional-scale flood susceptibility mapping in Kerala state, India

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Received: 20 December 2024 / Accepted: 21 May 2025

Published online: 04 June 2025

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Abstract

Floods in the southern Indian state of Kerala are the most frequent and deadliest types of natural disasters, leading to large economic losses and having killed over 410 people since June 2018. Despite Kerala's flood vulnerability and due to its intense annual rainfall, existing flood prediction approaches often fail to provide accurate and localized risk assessments. Various Machine Learning (ML) approaches offer promising results but there is a critical gap in applying these advanced ML algorithms along with systematized decision-making frameworks that would work well for Kerala's specific geographical and climatic conditions. This research aims to develop flood susceptibility maps for Kerala using a spatial flood database integrated within the ArcGIS interface. An efficient framework integrating Light Gradient Boosting Machine (LightGBM) learning models with the Analytic Hierarchy Process (AHP) is proposed to address this issue by enhancing flood susceptibility understanding and informed decision-making. The resultant map is validated using the Area Under Curve approach, which showed good accuracy of 0.948 and 0.763 for the LightGBM and AHP methods, respectively. The result indicates that the use of machine learning algorithm resulted better reliability of flood susceptibility maps. This study's findings apply to local authorities to maintain their preparedness and response efforts as well as minimize flood consequences.

Article highlights

- Novel combination of machine learning and expert knowledge systems creates more reliable flood susceptibility maps for Kerala.
- Study reveals that nearly one-fifth of Kerala's land area and a quarter of major roads lie in high-risk flood zones.
- Integration of landscape features helps identify safe zones and vulnerable areas for better disaster management planning.

Keywords Flood susceptibility · Analytical hierarchy process · Susceptibility · Remote sensing · GIS · Machine learning algorithm

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1 Introduction

Floods have been among the most frequently occurring natural disasters in recent years. In India, 45.64 million hectares of the total geographical area are liable to floods, as estimated by Rashtriya Barh Ayog (RBA) in its 11 th Five-year plan [1]. With typical geo-climatic characteristics that make it exceptionally susceptible to floods dramatically exposed during the catastrophic August 2018 floods, Kerala state faces exceptional flood vulnerability challenges [2]. At that time, the state had 758.6 mm of rainfall, 164 percent above normal levels, and the authorities released water from many dams as reservoirs became full [3]. The widespread flooding caused by it affected 13 districts and asked for the evacuation of 3.4 million people, signaling the need for better flood prediction management systems [4].

Flood Susceptibility Mapping (FSM) has emerged as an indispensable tool for mapping the zones vulnerable to floods, mainly when localized extreme rainfall events cause flash floods and landslides, as reported in recent cases in Kerala [5]. Rapidly, hyperlocal weather information for better flood early warning can fill gaps, challenges, and uncertainties in assessing flood susceptibility in emergencies and the following global warming [6]. A flood early warning system that uses hyper-local data to address the current gaps, challenges, and uncertainties in flood susceptibility is essential for protecting lives, properties, and livelihoods [7]. Such systems are necessary to safeguard lives, properties, and livelihoods in different regions impacted by other challenges [8]. Several limitations in current flood management approaches in India are that there are no detailed flood inundation maps of varying weather and river conditions and that the disaster management authorities develop no localized flood action plans [9]. Beyond immediate disaster response, accurate flood susceptibility assessment is important in risk assessment and knowledge for disaster preparation, land use planning, climate change adaptation, and economic impact mitigation [10]. The Geographic Information System (GIS) and Machine Learning (ML) methods successfully develop practical flood susceptibility assessment tools, and ensemble methods are more robust and accurate in predictions. These techniques for advanced studies in flood risk management and mitigation planning are helpful in any geographical context [11].

Several analytical approaches, combining ML models with the Analytic Hierarchy Process (AHP), are utilized to produce comprehensive flood susceptibility maps [12]. The AHP methodology sets up a hierarchical structure of the decision criteria based on the degree to which the experts consider the necessary factors [13].

Based on recent studies, including various factors such as Terrain Characteristics, Land Use/Land Cover (LULC), soil type, drainage density, rainfall distribution, and proximity to river networks, multi-source data integration into a GIS-based framework for flood hazard mapping is considered adequate [14]. While initial data collection for ML can be costly, the long-term benefits of more accurate predictions and reduced damage outweigh these costs. Integrating ML with GIS provides a robust framework for real-time flood hazard assessment, critical for proactive disaster management [15].

The combination of machine learning boosting techniques with AHP has been proven effective for flood susceptibility analysis in several recent investigations in Kerala. These methodologies have successfully developed flood-prone area maps using various algorithms and spatial data sources [16]. Through AHP techniques to identify a high flood risk of 1346.4 km² in the Idukki district, which was validated against map accuracy using the Receiver Operating characteristic-area Under Curve method (ROC-AUC) with a very high result of 0.91. By using different ML algorithms like Random Forest (RF), Gradient Boosting Machine (GBM), and Extreme Gradient Boosting (XGBoost), among this XGboost better performances with 92% accuracy and ROC-AUC values of 0.90 [17].

Combined ML boosting techniques with geospatial technology have been demonstrated effective in another flood susceptibility analysis in Kerala. The models analyzed 12 different flood conditioning factors to predict susceptibility levels in the Periyar River basin, using the Logistic Regression (LR), Support Vector Machine (SVM), Naive Bayes, RF, AdaBoost, gradient boost, and XGBoost models the model accuracies were measured using ROC-AUC, the RF shows better prediction rate with AUC 94 [18, 19]. The study by [20] in Slovakia's Topľa river basin presents a significant comparative analysis between Multi-Criteria Decision Analysis-Analytical Hierarchy Process (MCDA-AHP) and ML boosted tree models. The findings revealed that ML approaches achieved superior accuracy, with Binary Classification Tree and Boosted Regression Trees models showing accuracy rates of 87.70% and 91.42%, respectively, compared to 81.33% for MCDA-AHP.

Similarly, [21] demonstrated the effectiveness of machine learning techniques combined with remote sensing data for flood risk mapping in the Galikesh River basin of Northern Iran. The approach they took was innovative, using Google Earth Engine (GEE) to produce flood risk indices from Landsat 8 satellite imagery and the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM). They used Random Forest to infer that a specific conjunction

of Waterway and River Density, River Distance, Maximum One Day Precipitation, and Elevation accounted for about 68.27 percent of the total flood risk factors, even with Waterway and River Density being most responsible at 23.8 percent. As indicated on their flood risk mapping, the high and very high-risk areas were about 21 and 18 percent of the total study area, respectively.

The main objective of this research is to evaluate different flood susceptibility levels, including very high, high, moderate, low and very low susceptibility in Kerala state, by integrating with GIS-based AHP, remote sensing data, and ML techniques. The research enables flood risk management techniques to focus on the region's unique geographical and hydrological characteristics and contribute to minimize future flood damages. A well-established ROC curve assesses a test's overall performance. This framework can build flood susceptibility maps in areas with limited information or where preliminary assessments are necessary before presenting accurate flood mapping via traditional hydrological or hydraulic.

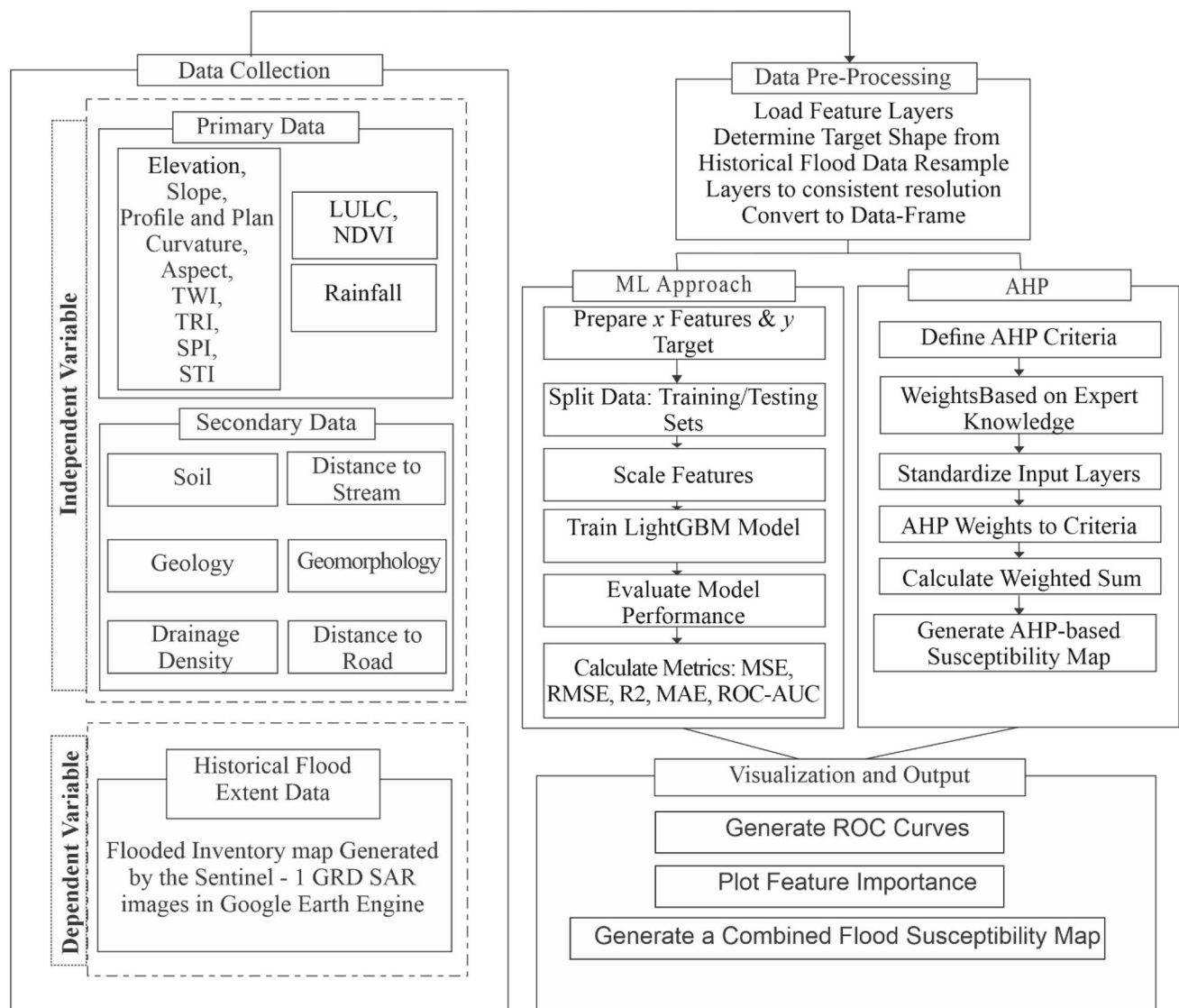


Fig. 1 Schematic diagram of flood susceptibility mapping

2 Methods and materials

This study employed a dual-model approach to flood susceptibility analysis, integrating both an ML-based predictive model and an Analytical Hierarchy Process (AHP) based model. The methodology consisted of several sequential steps, including data acquisition and preprocessing, feature selection, model development, and validation, as illustrated in Fig. 1. For this AHP, ten key factors that impact flood (TWI, Elevation, Slope, Precipitation, LULC, NDVI, Distance from rivers, Distance from roads, Drainage density, and Soil type) were processed by spatially homogeneous down sampling to target the exact resolution as historical flood amount. A light gum model was developed for the machine learning approach using a 70:30 train test split with binary classification objective, get boosting, 31 leaf nodes, 0.05 learning rate, and 0.9 feature fraction, and model performance evaluated as a function of MSE, RMSE, R^2 , MAE and AUC-ROC metrics after feature importance analysis to find out the most impacting flooding factors. Concurrently, the AHP approach incorporated expert knowledge by assigning specific weights to each factor shown in Table 1, standardizing input layers to a 0–1 scale based on their relationship with flood susceptibility, and calculating the Flood Susceptibility Index (FSI) using the weighted sum method ($FSI = \sum (W_i \times C_i)$ where W_i is the factor weight and C_i is the standardized value). The two approaches produced their respective flood susceptibility maps with probability values of 0–1, with higher values indicating a greater susceptibility, which, upon the fusion of both data-driven machine learning capabilities with expert knowledge integration through AHP, provides for a comprehensive assessment as results.

2.1 Study area description: Kerala state

Figure 2 presents Kerala state, situated in southwestern India (8°18' to 12°48' N and 74°52' to 77°22' E), encompassing an area of 38,863 km². The state exhibits varied physiographic features, encompassing coastal plains, midlands, and the Western Ghats Mountain range [22]. The proposed study examines the entire state, emphasizing its distinctive geographical and hydrological features that affect flood vulnerability [23]. The proposed research encompasses the following features that affect flood vulnerability: Elevation, Slope, Aspect, Profile and Plan Curvature, Distance to road, Distance to Drainage, Drainage Density, Terrain Ruggedness Index (TRI), Topographic Wetness Index (TWI), Stream Power Index (SPI), Sediment Transport Index (STI), LULC, Normalized Difference Vegetation Index (NDVI), Annual Rainfall, soil, Geomorphology and Geology.

2.2 Data collection methods

The Kerala flood susceptibility map was developed using remote sensing, and historical data. The Kallada River Basin analysis used GIS tools to classify sub-basins based on morphometric parameters and revealed that 51% of the study area is at risk of flooding [17, 24] applied two ML models, namely XGBoost and RF to the Vembanad Lake system and achieved high accuracy in flood susceptibility mapping.

The AHP in the Idukki district shows that more than 1346 km² is at high flood risk [25]. These studies have highlighted the need to integrate various data sources to develop adequate flood management strategies for Kerala state.

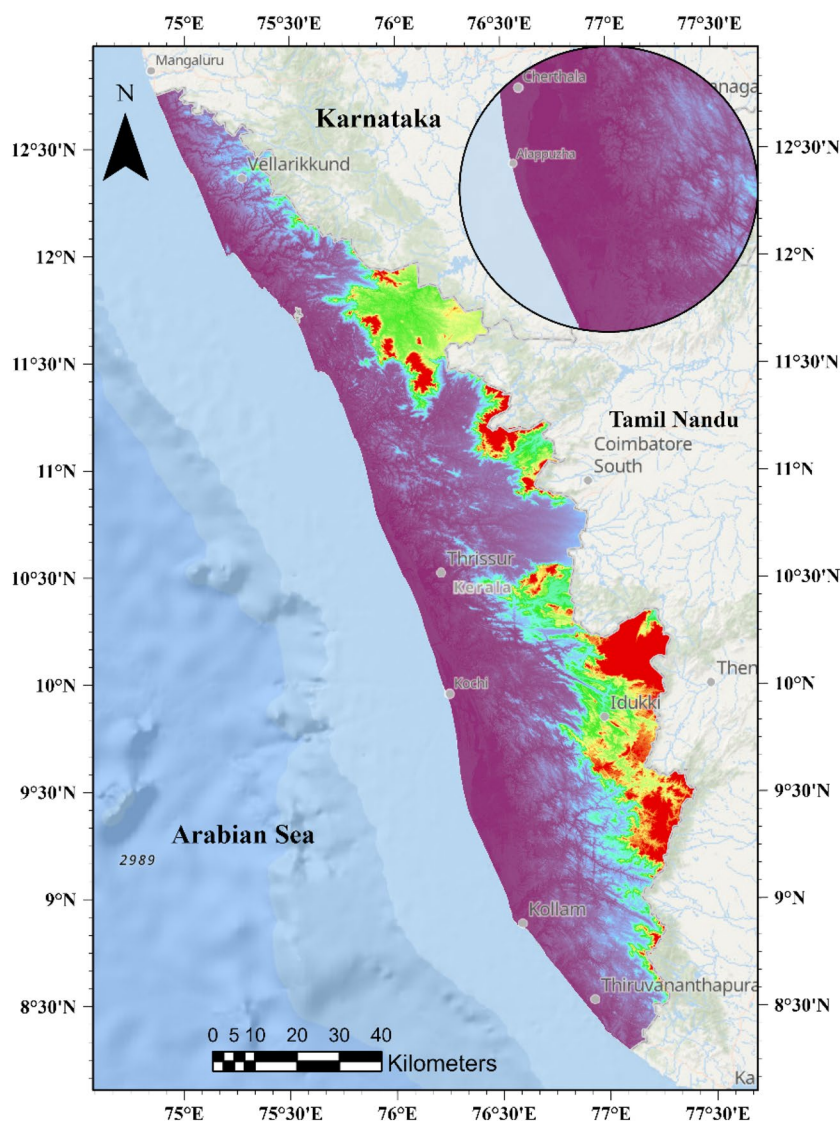
2.2.1 Remote sensing data

Continuous spatial information on land cover, topography, and flood extent is essential for the proposed investigations and is only available from remote sensing data. The data provided in this critical information allows for informed decision-making, enhanced disaster responses, and an improved understanding of the changing environment [26]. Since [27] used a combination of satellite imagery and derived products for the study area's geographical and temporal analyses, our research depended on a much different set of satellites, derived products, and time scales.

The spatial resolution of our topographic analysis is 30 m, and the NASA Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM). To derive these multiple critical parameters, we used the DEM to get the slope, aspect, profile, and plan curvature, Topographic Wetness Index (TWI), Terrain Ruggedness Index (TRI), Stream Power Index (SPI), and channelised Sediment Transport Index (STI). These derivatives give essential information about terrain's influence on water flow and accumulation [28, 29].

Table 1 Weightage of each factor and sub-factor of flood susceptibility parameters (FSP)

Flood causative criterion	Unit	Class	Susceptibility class ranges and ratings	Susceptibility class ratings	Weight (%)
Topographic wetness index (TWI)	Level	1.615–6.051	Very low	1	13
		6.052–8.127	Low	2	
		8.128–11.052	Moderate	3	
		11.053–14.921	High	4	
		14.922–25.679	Very high	5	
Elevation	m	– 48 to 236	Very high	5	12
		236.001–606	High	4	
		606.001–974	Moderate	3	
		974.001–1479	Low	2	
		1479.001–2685	Very low	1	
Slope	%	0.001–16.082	Very high	5	10
		16.083–32.165	High	4	
		32.166–48.247	Moderate	3	
		48.248–64.329	Low	2	
		64.33–80.411	Very low	1	
Annual rain	mm/year	73.562–99.933	Very low	1	15
		99.934–120.067	Low	2	
		120.068–135.44	Moderate	3	
		135.441–147.177	High	4	
		147.178–162.55	Very high	5	
LULC	Level	Water Body	Very high	5	9
		Agriculture land	High	4	
		Settlement	Moderate	3	
		Barren Land	Low	2	
		Vegetation	Very low	1	
NDVI	Level	– 0.281–0.102	Very high	5	6
		0.103–0.338	High	4	
		0.339–0.472	Moderate	3	
		0.473–0.583	Low	2	
		0.584–0.855	Very low	1	
Distance from river	m	1–0	Very high	5	14
		1–1113	High	4	
		1114–1579	Moderate	3	
		1580–2252	Low	2	
		2253–6601	Very low	1	
Distance from Road	m	1–1132	Very high	5	6
		1133–5063	High	4	
		5064–10,740	Moderate	3	
		10,741–17,720	Low	2	
		17,721–29,106	Very low	1	
Drainage density	m/Km	0.015–0.263	Very high	5	8
		0.264–0.512	High	4	
		0.513–0.761	Moderate	3	
		0.762–1.01	Low	2	
		1.011–1.259	Very low	1	
Soil type	Level	Silty clay	Very high	5	6
		Silty sediments	Moderate	3	
		Sandy clay sand	Low	2	

Fig. 2 Study area, Kerala State

Incorporated Hydro SHEDS data for hydrological parameters to analyse the drainage density and distance from stream network measurements derived from the SRTM DEM. Knowing these parameters is essential for understanding the mechanisms of flood propagation and calculating where the flood might overtop basins according to their closeness to the waterways.

Data from Bhukosh [30], the Geological Survey of India (for geology and geomorphology), and soil data from the National Bureau of Soil Survey and Land Use Planning integrated for geological factors. The information in these datasets is crucial to understanding how the subsurface conditions condition infiltration, runoff, and flood susceptibility.

Land Use/Land Cover (LULC) data from Esri and Sentinel-2 Land Cover products in the public domain are used to assess environmental parameters [31]. We also calculated the Normalised Difference Vegetation Index (NDVI) using Landsat 8 imagery accessed through Google Earth Engine to get the vegetation density and surface runoff. Rainfall data from CRU TS monthly high-resolution gridded multi-database choosing climate data between 2011 and 2020 were the meteorological component responsible for showing the direct influence of precipitation that leads to flood occurrence. In the context of an analysis of anthropogenic factors such as distance from road networks, as measured in the OpenStreetMap dataset, the analysis accounts for the influence of infrastructure development on river valley drainage patterns and, thus, on flood susceptibility.

This complete integration of 17 diverse parameters across topographic, hydrological, geological, environmental, meteorological, and anthropogenic categories helped to analyse the flood susceptibility factors of Kerala in an

all-inclusive manner. The multi-source remote sensing approach enabled capturing natural and human-modified landscape interactions as key drivers of which flooding vulnerability in the region is a product.

2.2.2 Historical flood records

The system generated a flood inventory across space and time using Sentinel-2 multispectral data (10 m resolution and 5-day revisit schedule) for all 2018 flood events within the GEE platform. The processing workflow comprised four steps. First, it performed atmospheric correction followed by cloud masking. Then, it applied spatial resampling and finally conducted temporal compositing [32].

2.3 Data collection challenges and solutions

Temporal misalignment poses a challenge to flood susceptibility mapping, as the time frames represented by historical records of flooding, rainfall, and land-use changes do not always coincide. The research learns from a comprehensive temporal harmonization framework that harmonizes historical data to solve historical inconsistencies and consistently analyses multiple time periods and data collection methodologies. First, align techniques were applied to synchronously combine disparate historical datasets. Second, sophisticated interpolation methods were applied to augment temporal data where measurements were absent. Finally, standardized five-year temporal windows were set up for consistent analysis of different living standards.

Different data layers, such as high-resolution STRM DEMs and Land use/Land cover, often have varying spatial resolutions, making their integration and analysis challenging [33, 34]. These integrated approaches ensure consistent and reliable spatial analysis across datasets of varying resolutions.

This study encountered several limitations in collecting SRTM DEM data, including coverage gaps in mountainous areas, poor coverage in steep terrain [35] and poor coverage of water bodies and coastlines. Variance errors and vegetation canopy interference cause vertical accuracy problems. Further limitations of the data quality result from technical limitations of the single-pass collection method and fixed resolution.

The research employs a comprehensive data preparation methodology to ensure quality and consistency. The void-filling methods, auxiliary DEMs, and interpolation methods are incorporated into this methodology to overcome SRTM data limitations. The approach maintains topographic accuracy, spatial consistency, and data continuity across the study area. The delta surface fill method successfully closes the SRTM DEM gaps, filling the research area with complete elevation as desired. Due to geographic consistency, the data is resampled to 30 m resolution and projected into UTM Zone 43 N. Compatibility with raster layers by standardisation at a 30 m scale that supports vector data conversion and helps establish a coherent analytical framework [36].

3 Data integration

In ArcGIS Pro, data integration to create a unified geodatabase. This process combines various data sources into a single, cohesive system and ensures continuous analysis and efficient data management throughout the study. The process involved several essential steps to integrate and consolidate information from various datasets.

3.1 Geospatial data standardization

All vector and raster datasets were converted to uniform UTM Zone 43 N projection and resampled to a 30-m spatial resolution. This essential data coordination phase provided spatial alignment of all data layers to facilitate exact overlay and analytical operations. This method also benefited from having a complete overview of the research area to search for changes in flood-related variables over time.

A comprehensive geographic database was created by combining topography, hydrologic, and weather data layers using the "Union" feature in ArcGIS Pro. This integrated database allowed for the simultaneous analysis of various factors that influence flood susceptibility. It addressed consistency issues across layers and more uniformly served as the key indicator for fetching and analysing multiple datasets simultaneously. The geodatabase consolidation improved the workflow efficiency for flood susceptibility modelling. Combining several datasets into a spatially consistent framework created a solid basis for subsequent flood dynamics analyses in the study area.

3.2 Spatial analysis tools

Spatial analysis tools in ArcGIS Pro integrate datasets to derive the significant metrics associated with floods. Furthermore, these methods were initially chosen and applied to obtain data necessary to further explore the theme of flood vulnerability modelling. This work used the Hydrology tools in ArcGIS Pro to delineate watersheds, calculate flow accumulation, and build stream networks of the SRTM DEM [37]. A fill tool allows us to remove spurious sinks from the DEM. The flow direction tool is then applied to find the flow direction. The flow accumulation tool calculates the accumulated flow to each cell, indicating potential flow paths and water accumulation points. Understanding the movement and distribution of water in the environment requires knowledge of specific hydrological characteristics, which are essential resources.

The researcher generated slope aspect curvature maps from a Digital Elevation Model using surface functionalities found in ArcGIS Pro Spatial Analyst. The slope function allows researchers to assess elevation patterns that control water speed and erosion intensity because these two factors play a crucial role in flood risk evaluation. The slope tool measures the steepness of elevation that affects flow velocities and erosion rates. It determines whether the elevation is convex or concave, affecting the water flow on the ground and its erosion risk (Curvature tool). Such topographical traits play a critical role in flood susceptibility.

The flood risk analysis methodology employs three essential ArcGIS Pro tools: the Surface Curvature tool, the Line Density tool (for drainage density calculation), and the Euclidean Distance. The Surface Curvature tool examines terrain morphology, identifying convex features (hills) and concave features (valleys). It is critical to know water flows and where erosion areas might be. The drainage density analysis gauges the complexity and size of the watershed drainage network by measuring the total length of all the streamways of the watershed about the drainage area. Higher drainage density is sometimes associated with higher flood risk due to faster water accumulation during rainfall.

Euclidean Distance calculates the straight-line distances from specific features, such as water bodies, drainage networks, and roads. These distance measurements identify areas and infrastructure at risk during a flood event. The analysis is performed at the administrative boundary scale, allowing a practical spatial scale of integrating and interpreting flood-related information.

The AHP was originally developed by [38] for solving any type of multi-criteria decision problem, using relative priorities assigned to each criterion towards achieving a decision-making procedure that presents the problem through a hierarchical structure [39]. AHP is a model with several criteria that are large and subjective marginal inputs. All inputs are converted to scores, and these are used to evaluate each criterion. AHP is found to be a powerful tool in the discipline of hazard management [40]. Steps in the AHP includes (1) Consideration of multi-criterion parameters; (2) Build a hierarchical order of chosen parameters; (3) Using subjective value assignment to establish relative importance and (4) Prioritization of 'alternates' by synthesis of ratings. Parameters in the pairwise comparison matrix (PCM) for flood susceptibility factors are given a hierarchical order, assigned values to signify their relative importance, and are presented in Table 2.

PCM is used to compute parameter weight, class weight, and the consistency ratio (CR) value. By calculating the CR using Eq. (1), the judgments made about the PCM of thematic layers and subclasses within those layers were validated:

$$CR = \frac{CI}{RI} \quad (1)$$

As presented in Table 3, the values of the Random Index (RI) used to compute consistency were taken from Saaty's standard, and the Consistency Index (CI) values were computed using Eqs. (2):

$$CI = \frac{(\lambda_{\max} - n)}{n - 1} \quad (2)$$

where $\lambda_{\max}$ is the maximum eigenvalue of the judgment matrix, and n is the number of criteria.

The pairwise comparison matrix of the 10 criteria AHP is analyzed for consistency leading to a $\lambda_{\max}$ of 10.730, which is 0.730 away from the ideal value of 10. This is a minimal deviation meaning there is a high level of consistency in the comparisons.

Consistency Index (CI) is calculated using Eq. 2 and it is found to be 0.0811. Consistency Ratio (CR) is found to be 5.5% as $RI = 1.49$ is the Random Index value specifically for a matrix with 10 criteria ($n = 10$), as shown in Table 3.

Calculations performed on the data set showed that annual rainfall had the greatest impact, having a weight of 15%, followed by the distance to rivers at 14% and TWI 13%. Elevation 12% and Slope 10% features were important while LULC comprised 9% of the susceptibility map. In all categories, each factor is placed into five ranks starting from low to

Table 2 Pairwise comparison matrix

Matrix	TWI	Elevation	Slope	Precipitation	LULC	NDVI	Distance from river	Distance from road	Drainage density	Soil type	normalized principal Eigenvector (%)
	1	2	3	4	5	6	7	8	9	10	
TWI	1										
Elevation	1	1	1	1	3	5	1	3	1	1	13.78
Slope	1	1	1	1	2	3	1	3	1	1	12.07
rainfall	1	1	1	1	3	1	1/2	1	1	1	9.90
LULC	1	1	1	1	3	2	2	3	1	1	13.45
NDVI	1/3	1/2	1/3	1/3	1	1	1/3	3	1	1	6.62
Distance from river	1/5	1/3	1	1/2	1	1	1/5	1	1	1	5.87
Distance from road	1	1	2	1/2	3	5	1	3	1	1	14.08
Drainage density	1/3	1/3	1	1/3	1/3	1	1/3	1	1	1	5.59
Soil type	1	1	1	1	1	1	1	1	1	1	9.32
	1	1	1	1	1	1	1	1	1	1	9.32

Table 3 Saaty's random index

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

high and given ratings from 1 to 5. The further analysis included distance-related factors such as the distance from rivers (14%) and roads (6%), in addition to drainage density (8%), NDVI (6%), and soil classification. This weighted hierarchy illustrates the significance of each factor concerning flood occurrence, facilitating a more accurate identification of areas susceptible to flooding, and thereby enhancing disaster risk management efforts.

3.3 Flood zone mapping

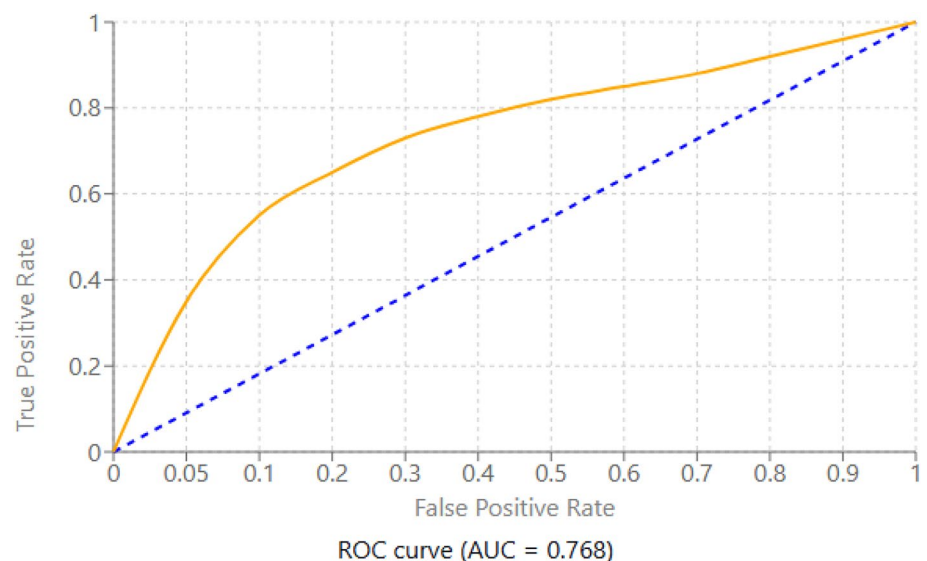
Figure 3 shows the ROC curve of the AHP method, which represents exceptional predictive accuracy in flood susceptibility mapping with an AUC of 0.769. It noted that this is not more than the random classifier (blue dashed diagonal line), illustrating that the model's has a robust discriminative capacity. It stabilizes upon attaining a true positive rate of 1.0, keeping the sensitivity perfect for any value of a false positive rate.

The AHP feature ranking analysis, illustrated in Fig. 4, presents a hierarchical distribution of the significance of flood-inducing elements, with error bars representing uncertainty ranges. The primary determinants are distance from the river and the TWI, accounting for approximately 14%, with precipitation and elevation closely trailing at around 12%. The high uncertainty ranges ($\pm 6\%$) on the leading factors, combined with diversity in expert evaluations, indicate there is diversity. For the intermediate tier, slope (10%) and drainage density (9%), LULC and NDVI show values of about 6–7%. The least influential factors, distance from road and soil type, have error bars overlapping significantly with the uncertainty ranges of other metrics.

Figure 5 illustrates the Flood Susceptibility Zone (FSZ) map of Kerala, obtained from the AHP method, highlighting a significant spatial dispersion of flood-hazard zones along the state's longitudinal extent (75°E to 77°E). The weighted overlay is used to generate FSZ from the 10 reclassified layers. The FSZ map incorporates five distinct categories: very low, low, medium, high, and very high. The map shows a distinct pattern with high-risk areas in the northeast and some coastal districts, and moderate-risk zones in central and southern Kerala. Extreme danger areas, isolated hotspots most notably near the coastline.

The research applies a multi-criteria decision analysis (MCDA) approach with ten critical parameters to develop the Flood Susceptibility Zone (FSZ) assessments. Pairwise comparison matrices serve to evaluate parameter weights while determining their corresponding normalised eigenvalues through this methodology. Distance from the river, TWI, and rainfall maintain the highest proportions towards flood susceptibility at 14.08%, 13.78% and 13.45%, respectively, in index analysis. The influence of NDVI and distance from the road on flood susceptibility is relatively minor since these factors have ratio values of 5.87% and 5.59%. The rating system utilises a 1–5 metric, determining parameter influence on flood

Fig. 3 ROC Curve of AHP method



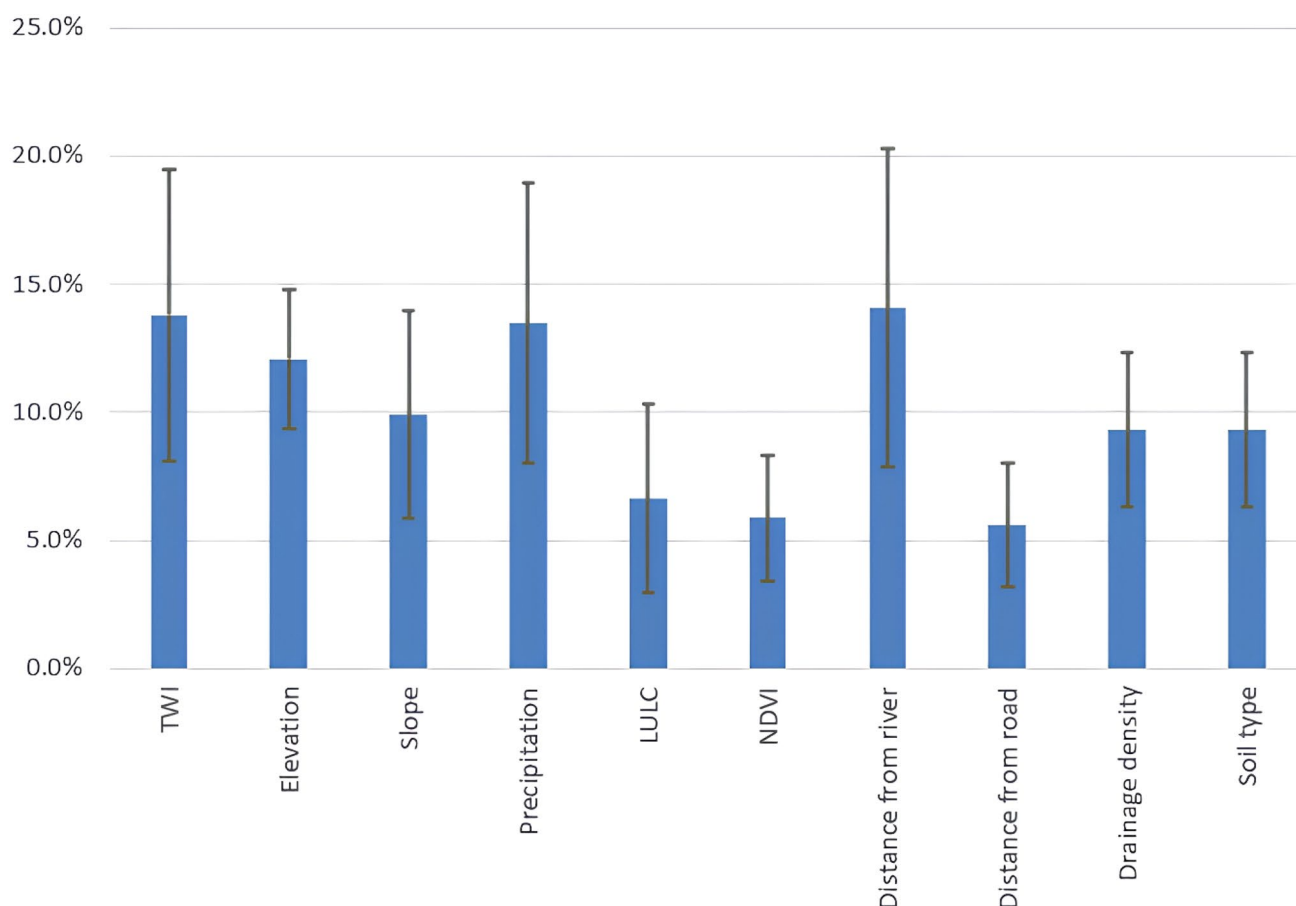


Fig. 4 AHP feature raking analysis

susceptibility levels. By applying their designated importance weights, the weighted linear combination method uses all rated parameters to produce the final FSZ value.

$$FSZ = \sum_{i=1}^n W_i^S \times R_i^S \quad (3)$$

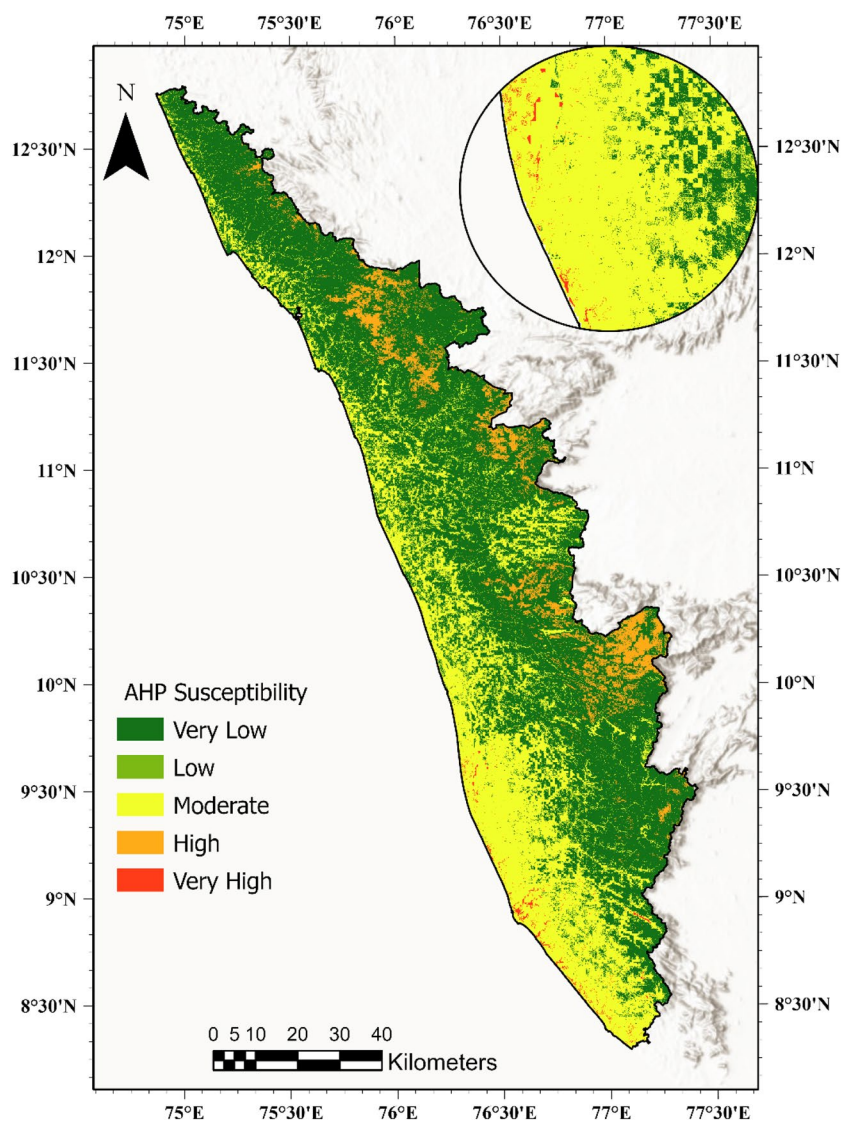
where W_i is the normalized weight of the parameter, and R_i is the rating of the parameter. FSZ value is 3.5325, which is high, as FSZ of 3.0–4.0 is considered as high flood susceptibility of the study area. The quantitative assessment of the hydrological and topographical factors as dominant factors in identifying areas prone to flooding highlights their importance, making this an essential contribution to flood risk management and spatial planning in the region.

4 Results and discussion

4.1 Evaluation of the GIS-based flood susceptibility model

Various performance indicators were used to assess the flood vulnerability model based on GIS for Kerala. The Receiver ROC analysis of the model resulted in an AUC of 0.948, which is the outstanding ability of the model to distinguish flood-prone and non-flood-prone areas. Overall accuracy of 94.8%, suggesting the projected and actual flood occurrences are correlated. The analysis of flood causative factors revealed a hierarchical influence pattern, as detailed in Table 1. The sensitivity analysis demonstrated that Annual Rainfall exhibited the highest influence ($\pm 15.5\%$), followed by Distance to the river ($\pm 14.8\%$) and Topographic Wetness Index (TWI) ($\pm 13.8\%$). This hierarchical pattern reflects the dominant role of precipitation and hydrological factors in determining flood susceptibility within the study area.

Fig. 5 Flood susceptibility zone by AHP



4.2 Machine learning model with integration of AHP

Model development is achieved using LightGBM, a gradient-boosting framework. The model is developed with binary classification, using gradient-boosted decision trees (GBDT) as the boosting type. To split the dataset into training (80%) and testing (20%) with reproducibility, the dataset was split with a random state of 42. The feature importance is calculated with AHP weight incorporated into the training process, which serves as a novel hybrid approach combining data-driven learning with expert opinion. Figure 7 shows ROC-AUC scores across the training dataset (0.9482) and testing dataset (0.9478) have very close consistency, with a difference of just 0.0004. In addition, the model prediction accuracy is further validated by using various error metrics. The model accuracy is validated using two error metrics, Mean Error (ME) which achieved 0.0017, and Root Mean Square Error (RMSE) of 0.0417. In addition, a Mean Absolute Error (MAE) of 0.0033 proved that mean absolute errors made in prediction are small on average. The transparent and interpretable AHP technique uses each factor's weight to represent its value in the estimation of flood susceptibility. Such a hierarchical system is very flexible, in that the weights assigned by stakeholders can be understood and changed according to local conditions or updated expert assessments.

The comparison shows the pros and cons of LightGBM and AHP flood susceptibility mapping. LightGBM has high accuracy and AUC for large-scale assessments, but its lower precision and recall suggest it could better forecast

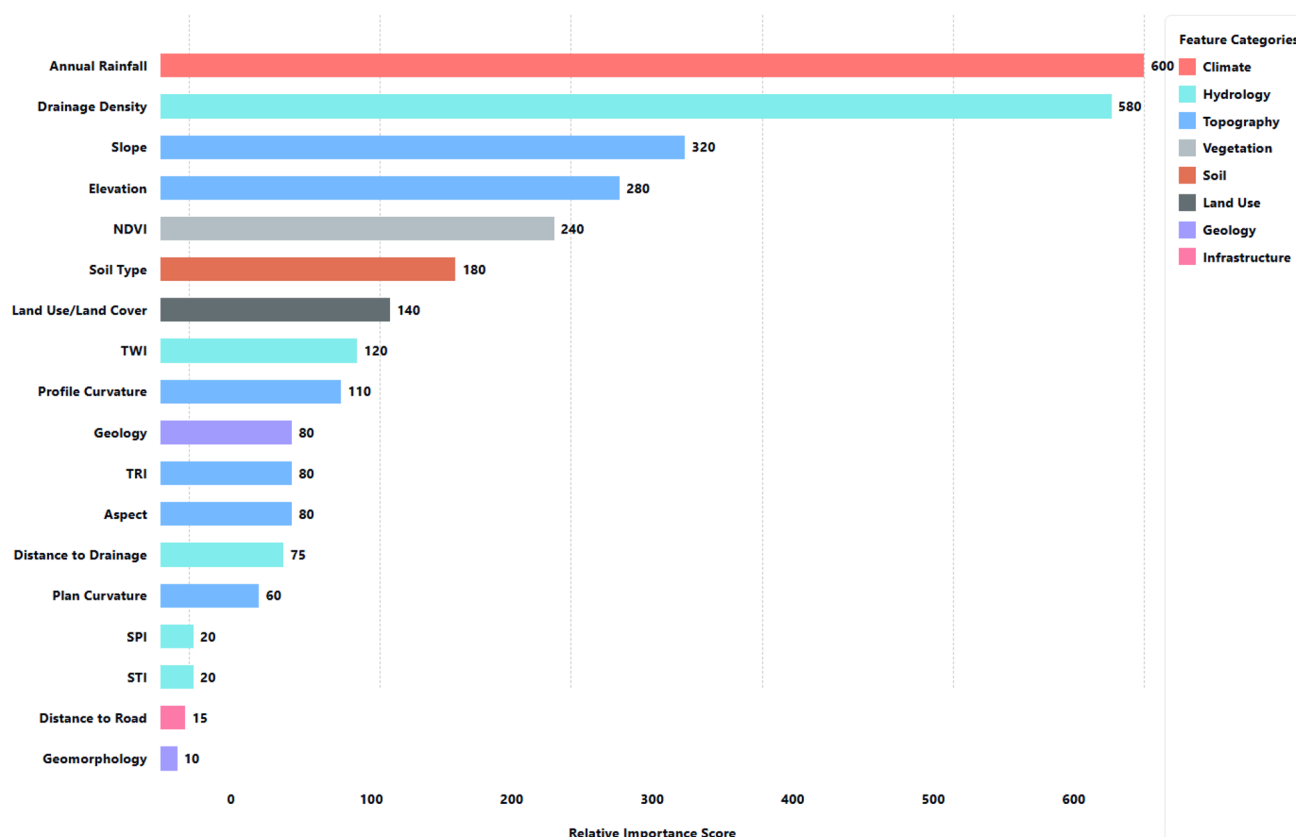


Fig. 6 Environmental feature importance by light BGM model

flood-prone locations. The AHP technique is more visible and interpretable, stressing the need for expert knowledge in assessing flood risk across terrain types.

Figure 6 distinctly prioritizes feature importance by LightGBM model for flood vulnerability assessment. The major predictor is further identified as annual rainfall, with a significance score of roughly 450. Kerala's drainage density and its elevation are highly correlated, exceeding scores of 350. Moderate (scores 250–300) in importance is the LULC and the elevation and precipitation interaction. The decrease in significance from topographic factors, such as TWI, altitude and soil, to the least impact factors is demonstrated by the model (Fig. 7).

4.3 Discussion of model performance

Several important insights regarding the performance characteristics of the model are obtained. The same ROC-AUC score between the training dataset (0.9482) and testing dataset (0.9478) shows that the predictive abilities generalize precisely to other data. For practical applications in flood risk management, this reliability is very useful.

The low MAE of 0.0033 is less than the RMSE of 0.0417 indicating that the predictions tend to be very accurate, but in some cases, predictions are average off because, as RMSE gives more weight to larger errors. In Fig. 8, The radar plot visualization shows a well-balanced model performance across different metrics, especially regarding error minimization (as shown by high 1-MSE and 1-MAE values above 0.99).

4.4 Implications for flood susceptibility map

The research determined that 18.5% of Kerala's area is very vulnerable to flooding (Fig. 9). There are considerable ramifications for zoning restrictions and land-use planning. The model's predictive performance (AUC 0.948) facilitates the advancement of more precise early warning systems. Infrastructure planning must consider that 23.5% of current main highways are in areas of high or extremely high risk. Resource allocation for flood management must prioritise the 18.3% of land categorised as having high or extremely high sensitivity.

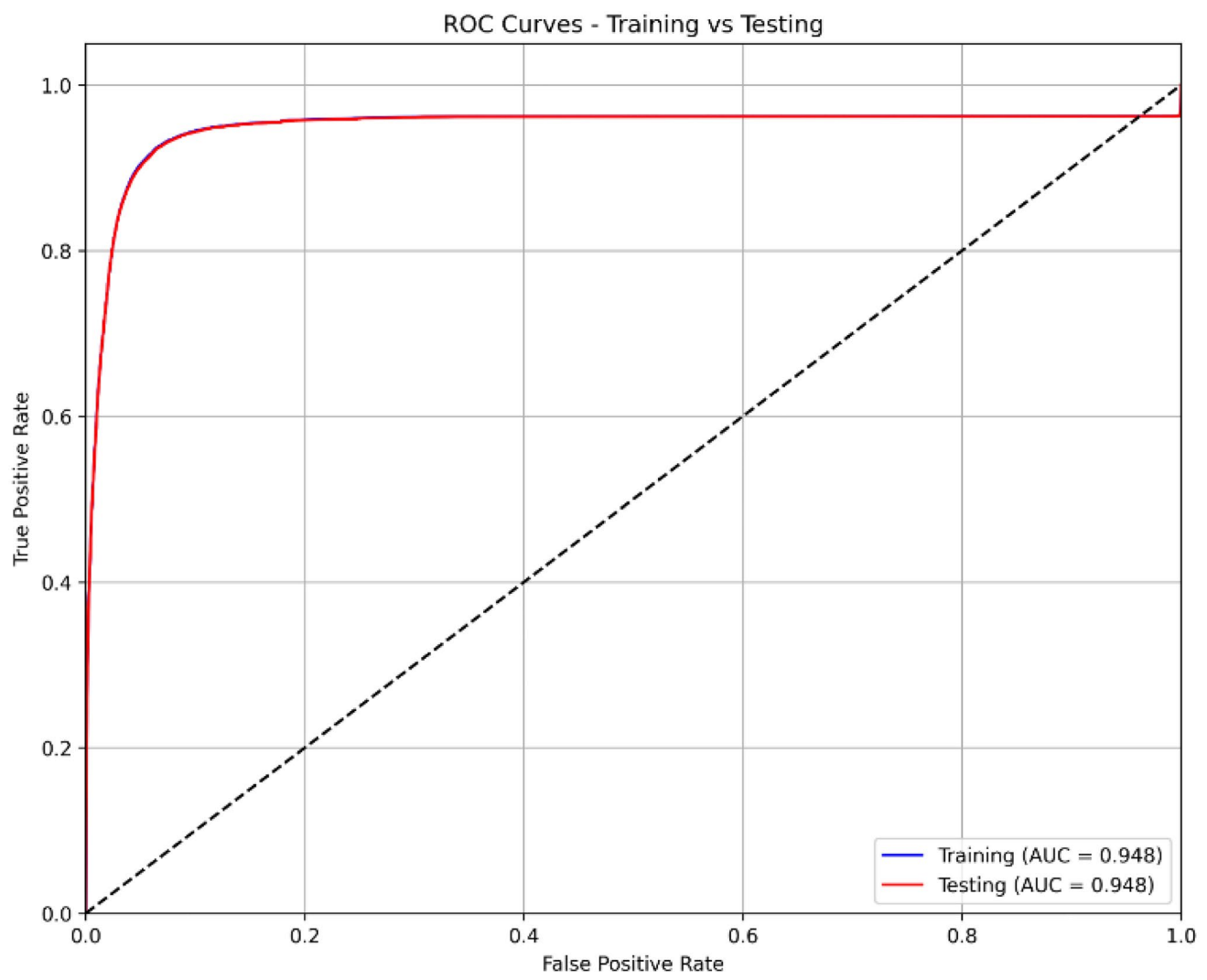


Fig. 7 AUC Roc curve for light BGM model

Fig. 8 Radar plot visualization for light BGM model

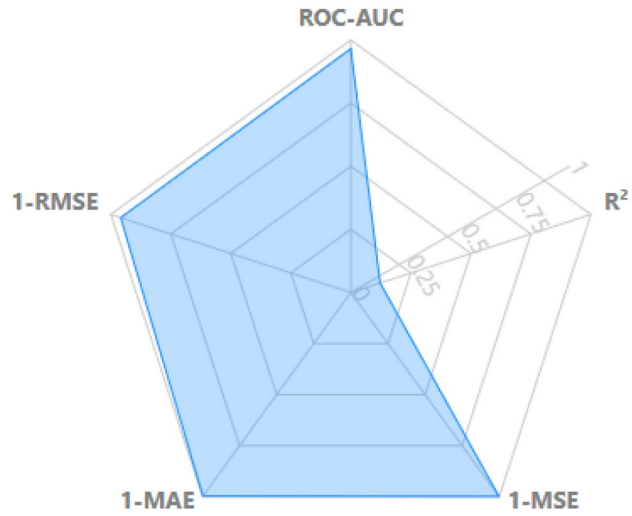
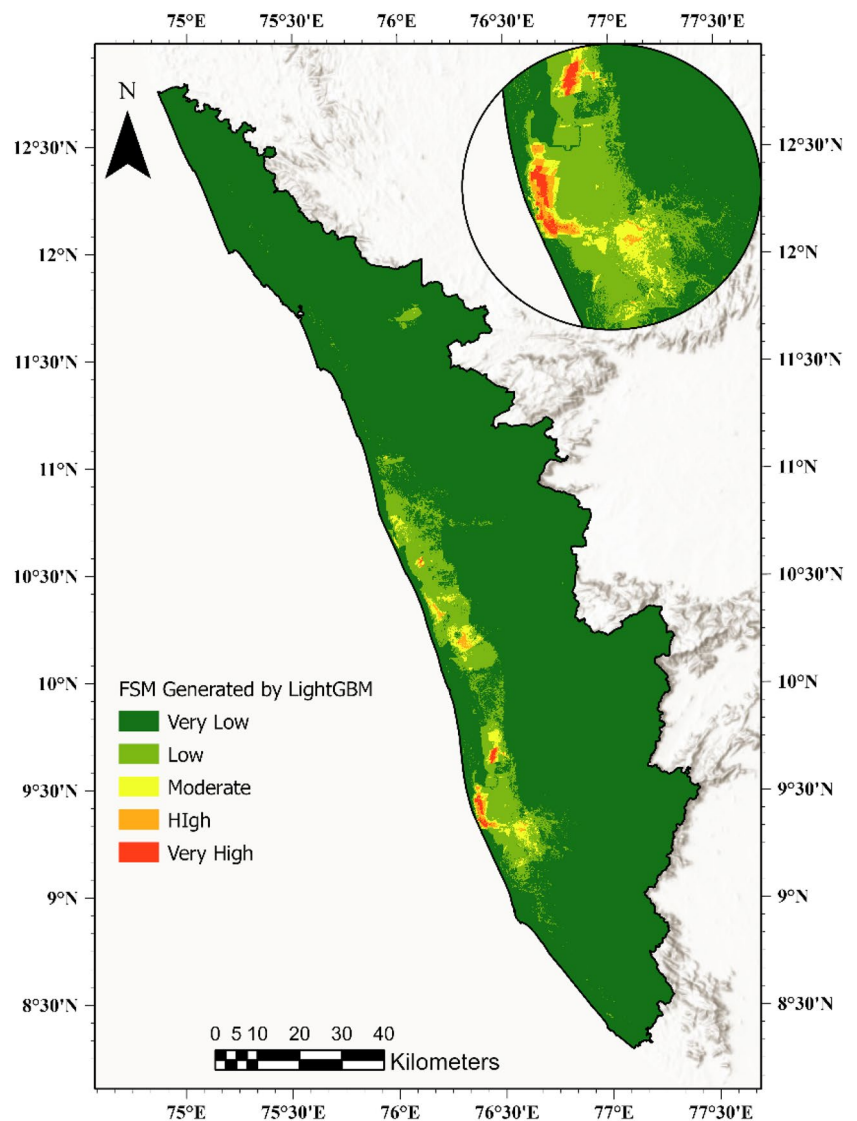


Fig. 9 Flood susceptibility map generated by LightGBM



The significant influence of Land use/Land cover (weight: 0.12) on flood susceptibility indicates that measures advocating for sustainable land use practices may significantly mitigate flood risk. The model's flexibility facilitates the integration of climate change scenarios, enhancing long-term flood risk management.

5 Conclusion

Primarily aimed at identifying the problem of lack of adequate flood risk assessment in the susceptible region of Kerala state of India, preliminary evaluation and development of a Geographical Information System (GIS) based flood susceptibility model using the combination of machine learning and expert knowledge is performed. The proposed model demonstrated an excellent predictive capability with an Area Under Curve (AUC) value of 0.948. A ranking of the most effective drivers of susceptibility to floods from analysis: rainfall patterns, elevation characteristics, and proximity to rivers. Our GIS-based approach was more accurate and computationally efficient than conventional methodologies. FINDINGS indicate that 18.3% of the Kerala area lies in highly vulnerable zones and that 23.5% of the major highways are in high or very high susceptibility zones.

This study then presents a comprehensive GIS framework integrating multiple datasets specific to Kerala's geography. This framework results in high-resolution (30 m) susceptibility maps by combining the Analytical Hierarchy Process (AHP)

technique with a machine learning method. The model validated with good robustness using the catastrophic 2018 Kerala floods data, which validates its reliability even in extreme cases.

Future research should include temporal flooding dynamics, coastal flooding mechanisms, and climate change scenarios to enhance predictive capabilities further. The model could be improved using real-time data through streams and higher-resolution Digital Elevation Models. Further model performance improvement can be improved through cross-validation testing and exploring more advanced machine learning algorithms methodologically.

This study has significantly contributed to flood susceptibility mapping in Kerala using GIS, based on a reliable, accurate, and effective GIS methodology that is much better than a conventional method. The results lead to continued research of the identified limitations to enhance the effectiveness of this framework in reducing flood hazards in Kerala.

Author contributions A: conceptualization, methodology, investigation, writing—original draft; B and D: data curation, formal analysis, visualization; C: supervision, writing—review and editing.

Funding The authors declare that there is no funding regarding this article.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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